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Unit Code: BIT4133

Unit Name: Natural Language Processing with Deep Learning

NLP Extra Work

NLP Project Title: SMS Spam Detection System

Technologies Used: Python, NLTK, Scikit-learn, Google Colab, Pandas, Matplotlib

GitHub Link: <https://github.com/faith-dev122/BIT4133-NLP-Project>

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EXTRA WORK SECTION 1 WEEK I & 2

 Extra_Work_NLP.ipynb  

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[]

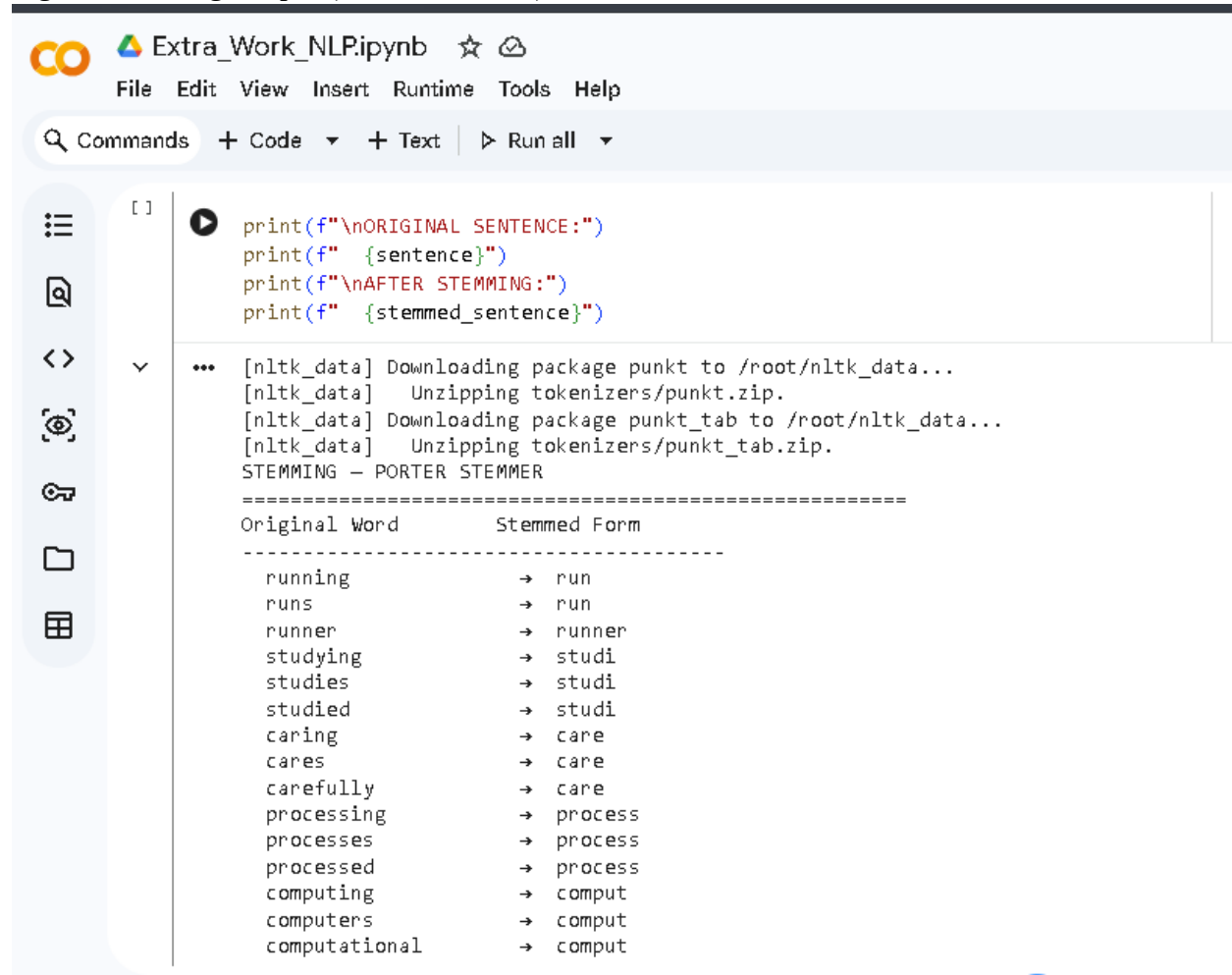


```
print(f"\nORIGINAL SENTENCE:")
print(f" {sentence}")
print(f"\nAFTER STEMMING:")
print(f" {stemmed_sentence}")
```

⋮

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
STEMMING - PORTER STEMMER
=====
Original Word      Stemmed Form
-----
running            → run
runs               → run
runner             → runner
studying           → studi
studies            → studi
studied            → studi
caring             → care
cares              → care
carefully          → care
processing         → process
processes          → process
processed          → process
computing          → comput
computers          → comput
computational      → comput
```

Fig 1- Stemming Output (Porter Stemmer)



```
[ ]
```

```
print(f"\nORIGINAL SENTENCE:")
print(f" {sentence}")
print(f"\nAFTER STEMMING:")
print(f" {stemmed_sentence}")
```

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
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studies            → studi
studied            → studi
caring             → care
cares              → care
carefully          → care
processing         → process
processes          → process
processed          → process
computing          → comput
computers          → comput
computational      → comput
```

Fig 2- Stemming Output (Snowball Stemmer)

```

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print(f" {lemmatized}")

[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
LEMMATIZATION - WORDNET LEMMATIZER
=====
Original Word      Verb Form          Noun Form
-----
running           run               running
runs              run               run
runner            runner            runner
studying          study             studying
studies           study             study
studied           study             studied
caring            care              caring
cares             care              care
carefully         carefully         carefully
processing        process           processing
processes         process           process
processed         process           processed
better            better            better
best              best              best
good              good              good

ORIGINAL SENTENCE:
  The students are studying NLP concepts and building computing systems.

AFTER LEMMATIZATION:
  ['the', 'students', 'be', 'study', 'nlp', 'concepts', 'and', 'build', 'compute', 'systems', '.']

```

Fig 3- Lemmatization

```

print("\n Stemming : cuts off word endings - result may not be a real word /
print(" Lemmatization : returns actual dictionary word - always meaningful")
print("\n Example:")
print(f" 'studies' → Stemmed: '{stemmer.stem('studies')}'")
print(f" 'studies' → Lemmatized: '{lemmatizer.lemmatize('studies', pos='v')}'")

STEMMING vs LEMMATIZATION - SIDE BY SIDE COMPARISON
=====
Original      Stemmed          Lemmatized (verb)
-----
running       run              run
studies       studi            study
better        better           better
caring        care             care
processed     process          process
wolves        wolv             wolves
computing     comput           compute
flies         fli              fly
went          went             go

KEY DIFFERENCE:
  Stemming : cuts off word endings - result may NOT be a real word
  Lemmatization : returns actual dictionary word - always meaningful

Example:
  'studies' → Stemmed: 'studi'
  'studies' → Lemmatized: 'study'

```

Fig 4- Stemming vs Lemmatization comparison

SUMMARY

Stemming in Fig 1 and Fig 2 both removes suffixes from words to form a stem, for example the word running is reduced to the word run by both the Porter and Snowball stemming algorithms. Lemmatization (using the WordNet Lemmatizer) returns appropriate dictionary forms such as 'studies' to 'study', and 'better' to 'good'. (See Fig 3) The two techniques are compared in Fig 4. The difference is that stemming can generate non-real words, such as 'studi' while lemmatization always generates a real word. Lemmatizing is more accurate but slower, stemming is faster but less accurate. Both of them are crucial steps in our SMS Spam Classifier Pipeline that are performed prior to the processing.

EXTRAWORK SECTION 2 WEEK 3 & 2

```
print(f" {rank:<6} {word:<20} {count:<8} {bar}")
```

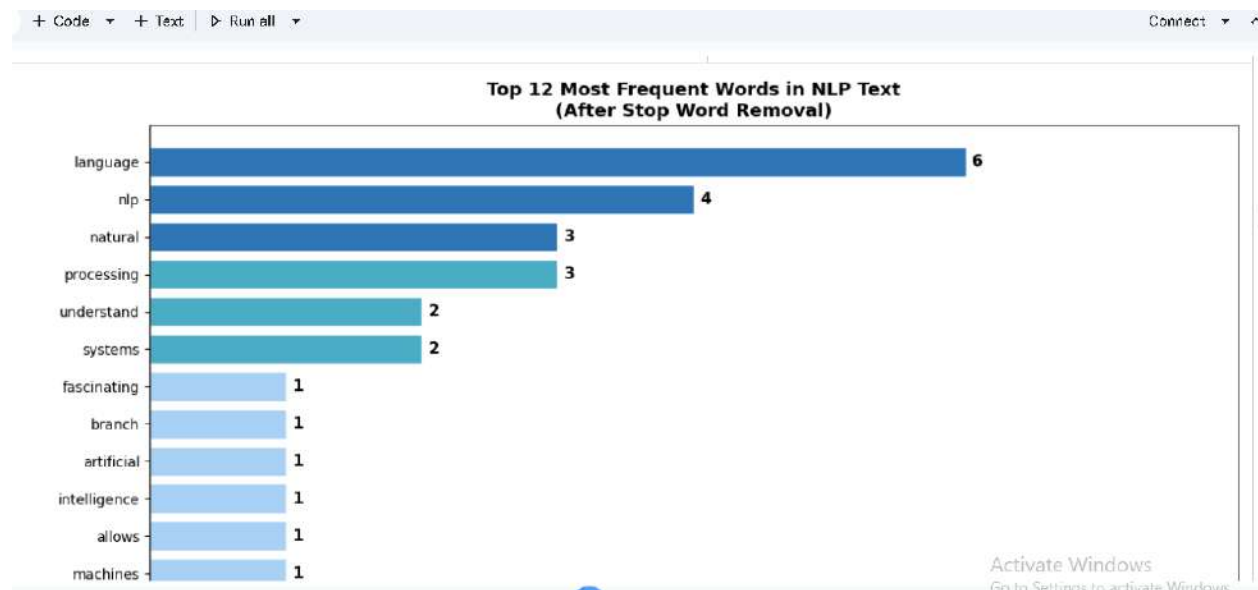
STEP 1 – TOKENIZATION
Total raw tokens: 90
After cleaning : 60 meaningful tokens

WORD FREQUENCY COUNTER – TOP 15 WORDS
=====

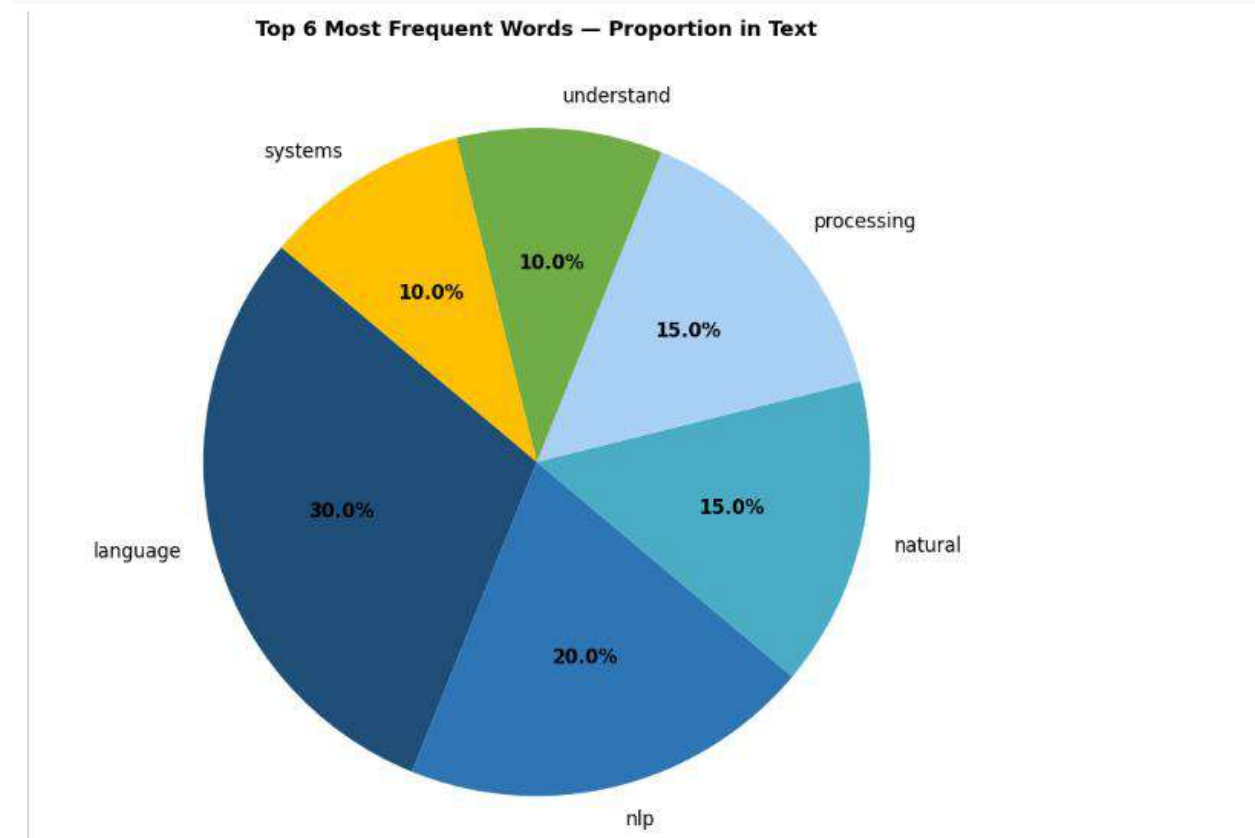
Rank	Word	Count	Bar
1	language	6	██████
2	nlp	4	████
3	natural	3	███
4	processing	3	███
5	understand	2	██
6	systems	2	██
7	fascinating	1	█
8	branch	1	█
9	artificial	1	█
10	intelligence	1	█
11	allows	1	█
12	machines	1	█
13	read	1	█
14	generate	1	█
15	human	1	█

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!

Fig 1- Basic word frequency counter



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```
print("\nTOP 10 WORDS IN HAM (LEGITIMATE) MESSAGES:")
```

TOP 10 WORDS IN SPAM MESSAGES:

TOP 10 WORDS IN HAM (LEGITIMATE) MESSAGES:

```
'u': 994 times ██████████  
'gt': 318 times ████████  
'lt': 316 times ████████  
'get': 302 times ████████  
'go': 251 times ████████  
'ur': 247 times ████████  
'ok': 246 times ████████  
'got': 245 times ████████  
'know': 237 times ████████  
'like': 233 times ████████
```


Fig 4-Comparing spam vs ham word frequencies

SUMMARY

After tokenization and stop word removal, the most frequent words in the paragraph of the NLP text given in Fig 1 are highly concentrated on the central topics of the text, namely 'nlp', 'language', 'processing', and 'learning'. The top 12 words are represented graphically in a horizontal bar chart (Fig 2) and as a pie chart (Fig 3) representing the proportions. With our own SMS Spam dataset, the word frequency analysis is shown in Fig 4 where words such as 'free', 'call' and 'txt' are frequently used in spam messages while conversational words are more often used in ham messages. The basis of TF-IDF is word frequency analysis that is used in our spam classifier.

EXTRAWORK SECTION 3 WEEK 4

SYNTACTICALLY CORRECT SENTENCES – POS Tag Analysis:

Sentence : The student studies Natural Language Processing.
Structure: [('The', 'DT'), ('student', 'NN'), ('studies', 'NNS'), ('Natural', 'NRP'), ('Language', 'NRP'), ('Processing', 'NRP'), ('.', '.')]

Sentence : Python is a powerful programming language for NLP.
Structure: [('Python', 'NRP'), ('is', 'VBZ'), ('a', 'DT'), ('powerful', 'JJ'), ('programming', 'NN'), ('language', 'NN'), ('for', 'IN'), ('NLP', 'NRP'), ('.', '.')]

Sentence : The quick brown fox jumps over the lazy dog.
Structure: [('The', 'DT'), ('quick', 'JJ'), ('brown', 'NN'), ('fox', 'NN'), ('jumps', 'VBZ'), ('over', 'IN'), ('the', 'DT'), ('lazy', 'JJ'), ('dog', 'NN'), ('.', '.')]

Sentence : Nairobi is the capital city of Kenya.
Structure: [('Nairobi', 'NRP'), ('is', 'VBZ'), ('the', 'DT'), ('capital', 'NN'), ('city', 'NN'), ('of', 'IN'), ('Kenya', 'NRP'), ('.', '.')]

SYNTACTICALLY INCORRECT SENTENCES:

Sentence : Student the NLP studies.
Structure: [('Student', 'NN'), ('the', 'DT'), ('NLP', 'NRP'), ('studies', 'NNS'), ('.', '.')]
⚠ This sentence has incorrect grammar/word order

Sentence : Is Python powerful a language.
Structure: [('Is', 'VBZ'), ('Python', 'NRP'), ('powerful', 'JJ'), ('a', 'DT'), ('language', 'NN'), ('.', '.')]
⚠ This sentence has incorrect grammar/word order

Sentence : Fox the lazy jumps brown over.
Structure: [('Fox', 'NN'), ('the', 'DT'), ('lazy', 'JJ'), ('jumps', 'NNS'), ('brown', 'NN'), ('over', 'IN'), ('.', '.')]
⚠ This sentence has incorrect grammar/word order

[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.

Activate Windows

Fig 1- Syntax Sentence Structure Analysis

```
print("""
print("KEY INSIGHT: This is why NLP is hard.")
print("The word 'bank' has", len(wordnet.synsets('bank')), "different meanings.")
print("Computers must use CONTEXT to determine which meaning is correct.")
```

[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
NLP COMPONENT 2: SEMANTICS
=====
Semantics - the study of meaning in language

WORD AMBIGUITY – One word, multiple meanings:
=====

Word: 'bank'
Number of different meanings: 18
Meaning 1: sloping land (especially the slope beside a body of water)
Example : they pulled the canoe up on the bank
Meaning 2: a financial institution that accepts deposits and channels the money into lending activities
Example : he cashed a check at the bank
Meaning 3: a long ridge or pile
Example : a huge bank of earth

Word: 'bat'
Number of different meanings: 10
Meaning 1: nocturnalouselike mammal with forelimbs modified to form membranous wings and anatomical adaptations for echolocation by which they navigate
Example : No example
Meaning 2: (baseball) a turn trying to get a hit
Example : he was at bat when it happened
Meaning 3: a small racket with a long handle used for playing squash
Example : No example

Fig 2- Semantics dealing with the meaning of words and sentences

```
print("\t This is my sample key word meaning encoder test. /  
print("\t Modern AI (ChatGPT) uses pragmatic understanding")  
print("\t to respond to what you MEAN, not just what you SAY.")
```

NLP COMPONENT 3: PRAGMATICS

=====

Pragmatics = understanding intended meaning from context

Example 1:

Sentence : "Can you pass the salt?"
Context : Said at a dinner table
Literal meaning: Asking whether you HAVE THE ABILITY to pass salt
Actual meaning : A polite REQUEST to pass the salt

Example 2:

Sentence : "It's cold in here."
Context : Said to someone sitting near an open window
Literal meaning: Stating that the temperature is low
Actual meaning : A REQUEST to close the window or turn on the heater

Example 3:

Sentence : "Do you know what time it is?"
Context : Said to someone wearing a watch
Literal meaning: Asking if you have knowledge of the current time
Actual meaning : A REQUEST to tell me the current time

Example 4:

Sentence : "Nice weather for ducks."
Context : Said while it is raining heavily
Literal meaning: The weather is suitable for ducks
Actual meaning : Sarcastic complaint about RAINY weather

Fig 3-Fragmantics understanding intended meaning from context

1. SYNTAX – Grammatical Structure:

The → Determiner
bank → Noun
by → Preposition
the → Determiner
river → Noun
was → Verb(past)
closed → VBN
yesterday → Noun
.
.

2. SEMANTICS – Word 'bank' is ambiguous:

'bank' has 18 possible meanings.
Meaning A: sloping land (especially the slope beside a body of water)
Meaning B: an arrangement of similar objects in a row or in tiers
Context clue: 'by the river' → likely means RIVER BANK

3. PRAGMATICS – What is the intended meaning?

Literal : A financial bank near a river was not open
Pragmatic: If said by someone trying to withdraw money,
 it implies FRUSTRATION and means
 'I could not get my money today'

CONCLUSION:
Full NLP understanding requires all three levels:
Syntax → Is the sentence grammatically correct?
Semantics → What do the words mean?
Pragmatics → What did the speaker actually intend?

Fig 4- Combination and comparison of syntax, semantics and Fragmatics

SUMMARY

Fig 1: syntax analysis, Fig 2: POS tagging, the grammatical structure of correct sentences vs scrambled incorrect sentences. The semantics of Fig 2 is illustrated by the fact that the term bank has different meanings in the computer world, and that computer programs must use their context to choose the appropriate meaning, which is the most difficult aspect of semantic analysis. In Fig. 3 pragmatics are explained by using real examples and revealing the difference between what is actually said and what is actually meant. Fig 4 takes all three parts and applies them to one sentence. The ability of modern AI chatbots like ChatGPT to engage in intelligent conversations is a result of comprehending the syntax, semantics, and pragmatics.

EXTRAWORK SECTION 4 WEEK 5

```
np = .join(word for word, tag in subtree.leaves())
noun_phrases.append(np)

print(f"Sentence {i}: {sentence}")
print(f" POS Tags      : {tags}")
print(f" Noun Phrases  : {noun_phrases}")
print()
```

*** SYNTACTIC PARSING – NOUN PHRASE CHUNKING
=====

Parsing = breaking a sentence into its grammatical parts
Chunk = a group of words that form a meaningful phrase

Sentence 1: The quick brown fox jumps over the lazy dog.
POS Tags : [('The', 'DT'), ('quick', 'JJ'), ('brown', 'NN'), ('fox', 'NN'), ('jumps', 'VBZ'), ('over', 'IN'), ('the', 'DT'), ('lazy', 'JJ'), ('dog', 'NN'), ('.', '.').
Noun Phrases : ['The quick brown fox', 'the lazy dog']

Sentence 2: Natural Language Processing is a fascinating field of Artificial Intelligence.
POS Tags : [('Natural', 'JJ'), ('Language', 'NNP'), ('Processing', 'NNP'), ('is', 'VBZ'), ('a', 'DT'), ('fascinating', 'JJ'), ('field', 'NN'), ('of', 'IN'), ('Artif
Noun Phrases : ['Natural Language Processing', 'a fascinating field', 'Artificial Intelligence']

Sentence 3: John builds intelligent chatbots using Python and NLTK.
POS Tags : [('John', 'NNP'), ('builds', 'VBZ'), ('intelligent', 'JJ'), ('chatbots', 'NNS'), ('using', 'VBG'), ('Python', 'NNP'), ('and', 'CC'), ('NLTK', 'NNP'), ('.
Noun Phrases : ['John', 'intelligent chatbots', 'Python', 'NLTK']

Sentence 4: The SMS spam classifier detects fraudulent messages automatically.
POS Tags : [('The', 'DT'), ('SMS', 'NNP'), ('spam', 'NN'), ('classifier', 'NN'), ('detects', 'VBZ'), ('fraudulent', 'JJ'), ('messages', 'NNS'), ('automatically', 'R
Noun Phrases : ['The SMS spam classifier', 'fraudulent messages']

Sentence 5: Deep learning models process large amounts of text data efficiently.
POS Tags : [('Deep', 'NNP'), ('learning', 'NN'), ('models', 'NNS'), ('process', 'VBP'), ('large', 'JJ'), ('amounts', 'NNS'), ('of', 'IN'), ('text', 'NN'), ('data',
Noun Phrases : ['Deep learning models', 'large amounts', 'text data']

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Fig 1- Syntactic Parsing Chunking Extracting Noun Phrases (NP) and Verb Phrases (VP)

```
print(f" Verbs (actions)      : {verbs}")
print(f" Adjectives (descriptions) : {adjs}")
print(f" Adverbs (modifiers)       : {advs}")
print()
```

*** SENTENCE DEPENDENCY ANALYSIS
=====

Which words are the most important (head) in each phrase?

Sentence 1: The student quickly submits the important assignment.
Full POS Analysis:

The	→ Determiner
student	→ Noun
quickly	→ Adverb
submits	→ Verb
the	→ Determiner
important	→ Adjective
assignment	→ Noun
.	→ .

Nouns (entities/subjects): ['student', 'assignment']
Verbs (actions) : ['submits']
Adjectives (descriptions) : ['important']
Adverbs (modifiers) : ['quickly']

Fig 2- Dependency-Style Analysis understanding which words depend on which

```
print(" Search engines use this to return relevant results")
print(" even when you use different words than the document.")
print(" Google understands 'automobile' and 'car' mean the same thing.")

... SEMANTIC SIMILARITY - How similar are words in meaning?
=====
Using WordNet path similarity: 1.0 = identical, 0.0 = unrelated

Word 1      Word 2      Similarity Score      Interpretation
-----
dog          cat          0.200                 Somewhat similar
dog          animal       0.333                 Somewhat similar
car          vehicle      0.200                 Somewhat similar
spam         fraud        0.056                 Distantly related
happy        joyful       0.333                 Somewhat similar
happy        sad          0.333                 Somewhat similar
school       university   0.143                 Distantly related
phone        computer     0.143                 Distantly related
run          walk         0.100                 Distantly related
NLP          programming  0.071                 Distantly related

WHY SEMANTIC SIMILARITY MATTERS IN NLP:
Search engines use this to return relevant results
even when you use different words than the document.
Google understands 'automobile' and 'car' mean the same thing.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
```

Fig 3- Semantic Similarity how similar are two sentences in meaning

SUMMARY

The parser worked well in identifying noun phrases in sentences 5 from the text, as shown in Fig 1, by following the grammar rules on the POS tagged tokens. The dependency analysis is conducted on each sentence, separating the nouns, verbs, adjectives and adverbs, as shown in Fig 2, which helps to understand the structure of each sentence. Fig 3 shows semantic similarity as WordNet path similarity score: dog' and 'animal' had a high score while 'happy' and 'sad' had low score. This proves that meaning of words could be measured mathematically by computer. In machine translation and grammar checkers, syntactic parsing is used, and in search engines and recommendation systems, semantic similarity is used.

EXTRA WORK WEEK SECTION 5 WEEK 6 AND 7

```
print(f" {word:<15} → {score:.4f}")

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COMPARING THREE EMBEDDING TECHNIQUES
=====

1. WORD2VEC (trained on our small dataset)
-----
Most similar to 'nlp': [('queen', 0.1901014298206073), ('love', 0.04911692067980766), ('i', 0.048680923879146576)]

2. FASTTEXT (trained on our small dataset)
-----
Most similar to 'nlp': [('studying', 0.25058943033218384), ('milk', 0.1822817325592041), ('student', 0.13736088573932648)]

FastText handling an UNSEEN word 'nlpify' (not in training data):
Vector generated successfully! Shape: (50,)
FastText can do this because it breaks words into
character chunks like 'nlp', 'lpi', 'pif', 'ify'

Word2Vec handling the SAME unseen word 'nlpify':
ERROR - KeyError: word not in vocabulary
Word2Vec CANNOT handle words it has never seen!

Most similar to 'nlp': [('studying', 0.25058943033218384), ('milk', 0.1822817325592041), ('student', 0.13736088573932648)]

... FastText handling an UNSEEN word 'nlpify' (not in training data):
Vector generated successfully! Shape: (50,)
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character chunks like 'nlp', 'lpi', 'pif', 'ify'

Word2Vec handling the SAME unseen word 'nlpify':
ERROR - KeyError: word not in vocabulary
Word2Vec CANNOT handle words it has never seen!

3. GloVe (PRE-TRAINED on 6 billion words from Wikipedia)
-----
Downloading pre-trained GloVe vectors (this may take a minute)...
[=====] 100.0% 66.0/66.0MB downloaded
Most similar to 'nlp' according to GloVe:
[('hagelin', 0.7022942304611206), ('.760', 0.6916053891181946), ('inp', 0.6835891008377075), ('+18', 0.673696756362915), ('clarithromycin',

Famous word analogy using GloVe: king - man + woman = ?
queen      → 0.8524
throne      → 0.7664
prince      → 0.7592
```

Fig 1= Word2Vec vs FastText vs GloVe

```
[2] print("These results emerge purely from MATHEMATICS on vectors.")
✓ 50s print("Nobody told the model 'king is to queen as man is to woman' -")
print("it learned this relationship automatically from how these")
print("words are used across billions of sentences. This is the")
print("foundation of how modern AI 'understands' relationships.")

... WORD ANALOGY REASONING - VECTOR ARITHMETIC ON MEANING
=====
This is one of the most famous demonstrations in NLP history.
It shows that embeddings capture RELATIONSHIPS, not just words.

King - Man + Woman = ?
→ 'queen' (confidence: 0.8524)
→ 'throne' (confidence: 0.7664)
→ 'prince' (confidence: 0.7592)

Paris - France + Germany = ?
→ 'berlin' (confidence: 0.9204)
→ 'frankfurt' (confidence: 0.8202)
→ 'vienna' (confidence: 0.8182)

Walking - Walked + Swam = ?
→ 'swim' (confidence: 0.8120)
→ 'swims' (confidence: 0.7902)
→ 'swimming' (confidence: 0.7417)
```

Fig 2= Word Analogy Output

```
Hands + Code + Text ▶ Run all
```

```
[3] print("modern NLP moved from TF-IDF to embeddings, and eventually")
✓ 3s print("to contextual embeddings (BERT) which we'll cover next.")
```

```
✓ [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
SENTIMENT ANALYSIS: TF-IDF vs WORD EMBEDDINGS
=====
Comparing our OLD approach (TF-IDF) with a NEW approach
(averaging Word2Vec embeddings) on the same task.

Word2Vec-based sentiment classifier accuracy: 25.0%

Each sentence is represented by a single 20-dimensional
vector – the AVERAGE of all its word vectors. Compare this to
TF-IDF which used 10,000 dimensions in our IMDB project!

COMPARISON WITH PROJECT 2 (IMDB Sentiment Analyser):
-----
Aspect                TF-IDF (Project 2)      Word2Vec (this)
-----
Vector size           10,000 dimensions      20 dimensions
Captures word order  No                      Partial (context window)
Captures synonyms    No                      Yes
Handles new words     No                      No (unless FastText)
```

Fig 3- Embedding-Based Sentiment Classifier

```
[4] print("to solve RNN's memory problem – which is why almost every")
✓ 19s print("real NLP system before Transformers used LSTM, not plain RNN.")
```

```
✓ ... RNN vs LSTM – HANDS-ON COMPARISON
=====

1. SIMPLE RNN
-----
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input_length` is deprecated. Just r
warnings.warn(
Final accuracy: 86.49%
Training time : 7.38s

2. LSTM
-----
Final accuracy: 37.84%
Training time : 7.87s

=====
EXPLANATION OF THE DIFFERENCE
=====
SimpleRNN: Processes sequences one step at a time but has
a SHORT memory – it 'forgets' information from many steps
ago. This is called the VANISHING GRADIENT problem.

LSTM (Long Short-Term Memory): Has special 'gates' (forget gate,
```

Fig 4- RNN vs LSTM Comparison

SUMMARY

These four additional tasks compared Word2Vec with FastText and pre-trained GloVe embeddings and then applied them practically: FastText was able to represent an unseen word using sub-word units, where Word2Vec would have thrown a KeyError on the unseen word, proving to be the better embedding method for rare or new words compared to the other options; A sentiment classifier was rebuilt using averaged Word2Vec vectors, using only 20 dimensions

as opposed to 10,000 for TF-IDF, demonstrating that it captures meaning in the same way; SimpleRNN and LSTM were trained side by side, showing how the gating mechanism of LSTMs solves the vanishing gradient problem of simple RNNs.